
THE EFFECT OF ANKLE BRACING ON PEAK MEDIOLATERAL GROUND REACTION FORCE DURING CUTTING MANEUVERS IN COLLEGIATE MALE BASKETBALL PLAYERS

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ABSTRACT

Cloak, R, Galloway, S, and Wyon, M. The effect of ankle bracing on peak mediolateral ground reaction force during cutting maneuvers in collegiate male basketball players. *J Strength Cond Res* 24(9): 2429–2433, 2010—The literature suggests that one-third of ankle injuries sustained during a collegiate basketball season are due to the poor execution of dynamic cutting movements, leading to increased mediolateral force being placed on the unstable ankle. The objective of the present study was to examine the effect of ankle bracing on peak mediolateral ground reaction forces (PMGRF) during sidestep cutting motions in collegiate basketball players. Ten male collegiate basketball players, with no previous history of ankle trauma, performed sidestep cutting motions past a static defensive opponent at speeds between 4.5 and 5.5 meters per second under 2 different conditions, with and without ankle bracing. Peak mediolateral ground reaction force was recorded (as a unit of body weight) for each subject in both bracing and control conditions. The application of an Aircast brace significantly ($P = 0.01$) reduced peak mediolateral forces during cutting maneuvers compared with no brace. Results suggest that the application of an Aircast ankle brace significantly reduces PMGRF during cutting maneuvers around a static defensive opponent.

KEY WORDS basketball, ankle support, ankle injury

INTRODUCTION

Ankle injuries are among the most common injury sustained by basketball players (19) accounting for over half (53.7%) of the total time missed (15). The sharp twisting or turning is a component of cutting and changing direction that is an integral part of basketball (12).

These movements are crucial to offensive strategies in basketball and are often accompanied with a rapid change in speed and/or direction to evade an oncoming defensive opponent (17). It has been suggested that one-third of ankle injuries sustained during a collegiate basketball season are due to the poor execution of dynamic cutting movements, leading to increased mediolateral force being placed on the unstable ankle (14).

Ankle braces have been reported as an effective method of restricting excessive mediolateral forces associated with injury (24,6), and thus the incidence of injury during the course of a basketball season (25). Research however remains equivocal on the effectiveness of ankle bracing during more realistic dynamic cutting movements. Further research has been recommended to assess the effectiveness of ankle braces during dynamic lateral cutting motions to help better understand the potential role ankle braces have at reducing forces and loads placed on the ankle-foot complex (5,20).

Comprehensive review of McClay et al. (13) of ground reaction force (GRF) among basketball players provided researchers with the first quantified data of forces acting on basketball players during typical game movements. It was noted that cutting maneuvers resulted in elevated mediolateral GRF during basketball-specific activities. Although an increased disposition to ankle injury due to the mediolateral forces was acknowledged, no formal investigation was undertaken into how to reduce such forces (13). The effect of the Aircast sport-stirrup on lateral shuffling movement at 80 to 90% of maximal speed exhibited a nonsignificant decrease in lateral peak impact force when compared with control groups (4). The researchers acknowledged that 80 to 90% of an individual's maximal speed on a lateral plane might not have been a sufficient enough speed to produce a large inversion moment, thereby not stressing the mechanical stiffness of the brace (4).

Clinicians have recognized the importance of reproducing game situation forces during the analyses of sidestep cutting dynamics (17). The introduction of a passive defensive opponent enhances the ecological validity of the research, with the resulting forces accurately reflecting the game environment (17,1). A direct problem to the reliability of

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results in such dynamic biomechanical studies is the concern with whether laboratory-based assessment of movements accurately reflects game movements (11); the introduction of a passive defensive opponent appears to contest this issue (17). The presence of a static defensive opponent produces a significant increase in peak mediolateral ground reaction force (PMGRF) during sidestep cutting movements, leading to a more game realistic force being applied to the subject during laboratory investigation (17). Despite basketball's widespread popularity, the biomechanics of the sport has received very little attention in the literature (13,23), with no study investigating the impact of a defensive opponent on GRF in basketball players during laboratory-based assessment. The aim of the present study is therefore to evaluate the effectiveness of an external ankle brace on reducing PMGRF in male recreational basketball players during sideward cutting movements past a static defensive opponent.

METHODS

Experimental Approach to the Problem

A randomized dependent group design was chosen as this would increase the likelihood that variations in data are due to the conditions rather than subject variability. Randomization of the trials reduces the effect of fatigue or familiarization, thereby reducing potential error or bias.

Subjects

Ten physically active, healthy, male University students (mean mass 794.2 ± 77.2 , height 171 ± 5.7 cm, and age 25 ± 5.2 years) participated in the present study. Participants had played basketball across a range of levels from collegiate to regional, and all exhibited appropriate skill levels to perform all activities during pilot testing conducted by a level 2 EBBA (English Basketball Association) coach. The study took place midway through the competitive season during a 2-week break in competition. All participants were screened for any significant lower extremity injury to the ankle or knee within the past 2 years (determined through verbal questioning by the researcher) to reduce any variables that may influence GRF (23,9). A male population was selected due to the significant differences between male and female mediolateral GRF during cutting and jumping motions (17,8,10). Ethical approval for the study was granted by the Research Centre for Sport, Exercise and Performance's Ethics Committee. Before any testing took place, participants were informed of the experimental risks and signed an informed consent document before the investigation.

Procedures

A Bertec 6090 force platform was used to collect PMGRF. The force platform was mounted flush with the surface of the floor within a university biomechanics laboratory. The frequency of response on the force platform was 500 Hz in all 6 components of ground reaction force; this sampling frequency has been previously reported as optimally compatible with dynamic cutting movements (23). The greatest

amount of PMGRF exhibited during contact time on the force platform was recorded for each trial. This value essentially represents the maximum mediolateral impact force on the ankle-foot complex (4). Cutting movements are typically executed during basketball at speeds between 4.5 and 5.5 meters per second (22) within the present study, a trial was deemed successful when executed within these limits. This was monitored by Brower light gates. The external ankle brace used during braced conditions was an Aircast semi-rigid ankle brace (Aircast, Inc, Summit, NJ) (Figure 1). The brace was chosen due to its perceived comfort and ease of application (22) and popularity among basketball players (23,16).

Each participant followed a uniform warm-up that consisted of progressively intensive running, multidirectional movement patterns, and dynamic stretches with detailed instruction given to them on the orientation of the session. All participants were tested under 2 ankle conditions: (a) control: no brace and (b) Aircast ankle brace (Aircast, Inc) in a balanced random order. The operational definition for assessing the dominant foot that would make contact with the force plate was done by asking the participants which leg they would use to kick a ball as far as possible (7,8). Each participant completed 3 trials for each condition with approach speeds falling between 4.5 and 5.5 milliseconds. A 10-minute rest



Figure 1. AirSport™ Ankle Brace (Aircast, Inc, Summit, NJ, USA).

was given between each experimental condition to reduce fatigue contribution to decreased GRF (27). After participants were instructed, they began their approach to the force platform. The authors decided on using a skeleton as the static defensive opponent to reduce collision risk. The skeleton was situated 20 cm behind the force platform and in line with the original direction of motion (Figure 2) to induce realistic cutting motion when participants made initial contact with the force platform.

Cutting angles were measured at 45° from the original movement direction in accordance with values typically observed during game situation (12,18). The angle was measured from the center of the force platform with a corresponding line marked clearly (using tape) so it could be seen by participants (Figure 3). On striking the force platform, participants were required to continue running for 2 m along the 45° angle into the boxed off area where they would receive a chest pass from a feeder.

A trial was deemed successful if the initial run-up speed fell within the prescribed range, the entire foot landed on the force platform, and the first step after the cutting maneuver fell on the 45° angle. Finally, the participant was required to successfully catch the chest pass within the boxed off area. Researchers stationed in different positions, visually

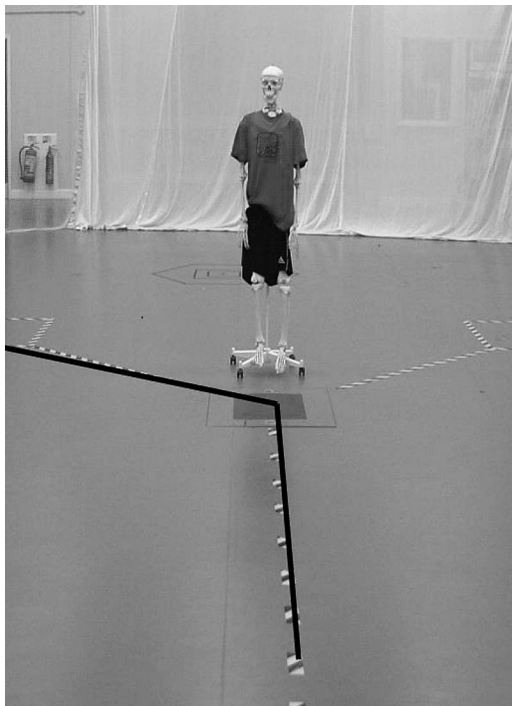


Figure 2. Illustration of subjects starting position and original line of motion. Static defensive opponent is placed to encourage successful cutting maneuver and in University kit to increase consistency with basketball environment.

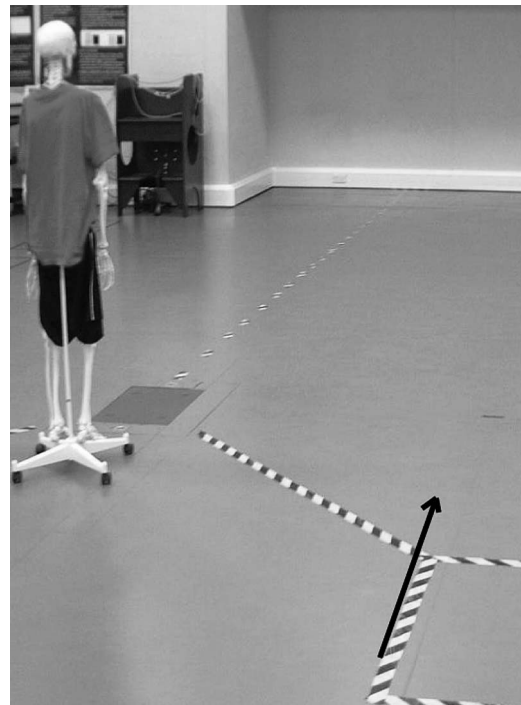


Figure 3. Represents the 1 × 1 meter boxed off area where the subject will receive a chest pass from a stationary feeder (illustrated by the thick black arrow).

controlled all trials, and instructed them to be repeated if deemed incorrect.

Statistical Analyses

The PMGRF data for each trial were analyzed using an independent *t*-test. An alpha level of $p \leq 0.05$ was used for all statistical analyses.

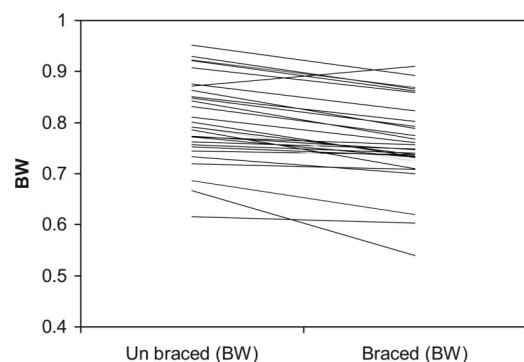


Figure 4. Group data for PMGRF under ankle brace and no ankle brace conditions. Forces are measured as a unit of body weight (BW).

RESULTS

Lower PMGRF values during cutting maneuvers were exhibited when participants wore an Aircast ankle brace compared with no brace (Figure 4). Independent t -test indicates statistically significant ($t_{58} = -2.93$, $p < 0.005$, $r = 0.36$) difference between PMGRF values when wearing an Aircast ankle brace (mean $0.8 \text{ BW} \pm 0.06$) compared with no brace (mean $0.75 \text{ BW} \pm 0.05$), with a medium effect size (3).

DISCUSSION

The present study results indicate that the application of Aircast ankle brace significantly ($p = 0.01$) reduces PMGRF during cutting maneuver around a static defensive opponent. Ankle sprains often occur due to poor foot positioning on initial impact with the ground and the resulting mediolateral force inverting the foot, causing injury (27). As with previous research, the application of an external ankle brace appears to increase stabilization to the rapidly induced mediolateral forces compared with wearing no brace at all (6). However, the present research provides a considerably more dynamic testing procedure, moving away from previous passive single leg stance measures (6). The effectiveness of any ankle brace at reducing mediolateral force can only be established during dynamic game-specific activity (20).

Due to the present study being the first to report the effects of external ankle bracing on peak mediolateral GRF when performing a cutting maneuver around a defensive opponent, comparisons are limited. However, raw mediolateral GRF data demonstrated by participants during the present research show similarities with studies of McClay et al. (13). Although there is similarity of results with McClay et al., the present study has greater ecological validity as the data bear closer resemblance to that of professional basketball players (17).

The results of the present study are in direct contradiction to Cordova et al. (4) research that reported no significant difference in PMGRF with the application of an Aircast external ankle brace. They acknowledged that insufficient force on the subject's ankle was created during testing (4). The present study identified this limitation in experimental protocol and therefore introduced a passive defensive opponent to increase mediolateral GRF to that of game-specific values as demonstrated by McLean et al. (17). The use of light gates to standardize approach speeds has also increased the validity of results obtained in the present study (21).

The present research does acknowledge that although all participants were screened for high-top basketball shoes, due to their effect on ankle stability in basketball (2), the results obtained still may have been affected by differences in footwear chosen by each individual that ultimately leads to problems when comparing group data means (26).

This study was designed to determine if ankle bracing reduces PMGRF during game-specific cutting maneuvers. The data provide evidence that the application of a semi-rigid

ankle brace significantly reduces PMGRF compared with those individuals who did not wear the ankle brace. It was also observed that the testing protocol used produced sufficient values in recreating basketball-specific movements with the implementation of a static defensive opponent. Although these results are encouraging, more studies are needed to understand the mechanisms by which the application of a semi-rigid ankle brace decreases PMGRF during dynamic cutting maneuvers on either or both ankles simultaneously. Further research is still required to assess the possible performance implications such as decreased running speed, agility, and vertical jump height through prolonged ankle bracing over the course of a basketball season.

PRACTICAL APPLICATIONS

The results in the present study provide some support for clinicians who advocate the use of prophylactic ankle support in healthy subjects during participation in basketball. The use of an ankle support is a potentially good preventative measure helping to reduce the mediolateral forces during cutting maneuvers. However, the long-term effects of wearing ankle supports are not known, and central to this is whether the body becomes reliant on this external structure to provide support rather than internal proprioception.

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